# Body Fat Percentage (BFP) Formulas

U.S. Navy Method (Hodgdon & Beckett, Naval Health Research Center, 1984) and the BMI Method

## Required Measurements

- **Waist** – Measured at navel level (horizontal) for men; at the narrowest point for women. Do not pull the stomach inward.
- **Neck** – Measured just below the larynx, with the tape sloping slightly downward toward the front. Avoid flaring the neck outward.
- **Hips** – (Women only) Measured at the largest horizontal circumference.
- **Height** – Standing height, required for all Navy method calculations.

## Navy Method – Men

### USC Units (inches):

$$BFP = 86.010 \times \log_{10}(waist - neck) - 70.041 \times \log_{10}(height) + 36.76$$

### SI Units (centimeters):

$$BFP = 495 / [1.0324 - 0.19077 \times \log_{10}(waist - neck) + 0.15456 \times \log_{10}(height)] - 450$$

## Navy Method – Women

### USC Units (inches):

$$BFP = 163.205 \times \log_{10}(waist + hip - neck) - 97.684 \times \log_{10}(height) - 78.387$$

### SI Units (centimeters):

$$BFP = 495 / [1.29579 - 0.35004 \times \log_{10}(waist + hip - neck) + 0.22100 \times \log_{10}(height)] - 450$$

Note: These formulas provide only an estimate of body fat percentage, based on population assumptions intended to be broadly applicable. For more precise measurements, methods such as bioelectrical impedance analysis (BIA) or hydrostatic (underwater) density testing are recommended.

## Fat Mass and Lean Mass

$$\text{Fat Mass (FM)} = BF \times \text{Weight}$$

$$\text{Lean Mass (LM)} = \text{Weight} - FM$$

## BMI Method

An alternative estimate of body fat percentage can be derived from BMI (Body Mass Index), which is calculated from a person's height and weight. Given BMI and age, the following formulas estimate body fat percentage:

| Group | Formula |
|---|---|
| `Adult Men` | `BFP = 1.20 × BMI + 0.23 × Age − 16.2` |
| `Adult Women` | `BFP = 1.20 × BMI + 0.23 × Age − 5.4` |
| `Boys` | `BFP = 1.51 × BMI − 0.70 × Age − 2.2` |
| `Girls` | `BFP = 1.51 × BMI − 0.70 × Age + 1.4` |

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Variables: BFP = body fat percentage; BF = body fat as a decimal fraction (BFP ÷ 100); waist, neck, hip, height measured in the units matching the chosen formula (inches for USC, centimeters for SI); log■■ = base-10 logarithm.